

EQUILIBRIUM-BASED CONTROL AS A DESIGN PHILOSOPHY FOR COMPUTING SYSTEMS

A design philosophy for computing systems — a measured result, a hardware corollary, and a general principle

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Abstract

Modern computing has adopted maximisation-based control as the default across the stack — chip design, operating systems, workload schedulers, cloud orchestration, building HVAC. This produces well-documented inefficiencies: high idle power, thermal cycling under load, resource waste, opportunistic peak-seeking that rarely aligns with what an outcome requires. Homeostatic control theory has understood the alternative for decades, but it has not been articulated as a unified design philosophy applicable across the whole compute stack.

This paper argues that equilibrium-based control, applied consistently from silicon to fleet, produces significant efficiency gains. It presents three pieces of evidence: a measured energy result showing a cascade routing pattern using a governance overlay as first-stage filter delivers 277 mJ per correct decision against a 774 mJ baseline (0.358 ratio, idle-subtracted CPU package power) compared to a monolithic small transformer on the same workload; an analytic model showing that a hardware architecture built on the same principles is consistent with a claimed 2.74 TOPS/W efficiency, subject to one load-bearing assumption (the sustained-fraction advantage of homeostatic thermal control) that requires FPGA thermal characterisation under continuous inference load to validate; and a general framing showing the principle applies wherever a goal-directed system operates under resource constraints with variable demand.

The paper explicitly does not claim to originate the equilibrium principle. Homeostasis has been named since Cannon (1932). Negative feedback control has structured engineering since Wiener (1948). Aristotelian virtue ethics and the Buddhist middle way both articulate variants at civilisational scale. What this paper contributes is the specific observation that modern computing has quietly adopted the opposite philosophy, that the resulting inefficiency is systematic and measurable, and that a coherent design alternative exists and can be tested.

1 The maximisation default

Modern computing systems, at almost every layer of abstraction, are designed to maximise something. A CPU boosts to peak frequency when work arrives. A

GPU pushes to thermal limits under load. An operating system scheduler opportunistically bursts to complete work as quickly as possible. A cloud orchestrator autoscales aggressively to meet demand spikes. A database runs background maintenance whenever resources are available. A HVAC system runs the compressor at full power when temperature drifts.

Each of these decisions, made independently, appears reasonable. Taken together, they produce a stack in which every layer opportunistically maximises its own metric. On workloads where peak-oriented operation is not required — which is most workloads outside sustained training runs — the aggregate result is that the system consumes substantially more resources than the outcome requires.

This paper argues that the maximisation default is the wrong control philosophy for the environment computing operates in today. The argument does not rest on any single technical result. It rests on the observation that maximisation-based design assumes conditions that stopped being true decades ago: continuous demand, scarce compute, unlimited energy, and dedicated infrastructure. These assumptions increasingly fail for important classes of modern workload. For much consumer and enterprise inference, compute is abundant, demand is bursty, energy is expensive and constrained by climate obligations, and infrastructure is shared across variable workloads. There remain workloads (sustained training runs, latency-critical real-time systems, dedicated high-performance computing) where the older assumptions still hold; the argument here does not apply uniformly to every case. Under current conditions, maximisation-based control produces systematic waste that could be recovered by different design choices.

The alternative is not new. Homeostatic control, sustained-equilibrium targeting, and predictive rather than reactive management have been well-understood in biology and control theory for most of a century. What is new is the observation that these principles have not been adopted as a unified design philosophy for computing systems, and that doing so would produce meaningful efficiency gains at every scale from personal devices to national infrastructure.

2 Precedent and positioning

The equilibrium principle in the sense used here has three main precedents.

Biological homeostasis. Cannon (1932) named the property by which biological systems maintain stable internal conditions in the face of external variation. Body temperature, blood glucose, blood pressure, and dozens of other variables are actively regulated toward set points rather than allowed to drift or maximise. When the set point is reached, the regulatory drive quiets. Perturbations are corrected. The result is a system that maintains function across a wide range of external conditions using far less energy than a maximisation-based alternative would require.

Control theory. Wiener (1948) and subsequent developments in classical and modern control theory formalised the mathematics of feedback and set-point regulation. Modern control systems, from industrial process control to aircraft

autopilots, are built on these foundations. What has not happened is systematic application of the same principles to consumer and enterprise computing, where control decisions are still largely reactive and threshold-based rather than predictive and equilibrium-oriented.

Philosophical antecedents. Aristotelian virtue ethics articulates the golden mean as the correct principle of action, holding that virtue lies between deficiency and excess. Buddhist ethics articulates the middle way as the path between indulgence and denial. Both traditions contain the general insight that maximisation of any single dimension produces pathology, and that sustainable operation requires targeting an equilibrium rather than an extreme.

This paper contributes none of these. What it contributes is the specific observation that modern computing has adopted the opposite philosophy without noticing, and the specific proposal that adopting equilibrium-based control across the stack would produce measurable efficiency gains. The technical contribution is a measured result on real silicon, an analytic model of a hardware architecture built on the principle, and a general framing that lets engineers in specific domains recognise their own problem in the framework and apply it.

The evidence in this paper is organised at three explicit levels, and readers should hold each at the epistemic status it warrants.

- **Level 1 (measured):** The cascade energy result in Section 4. Real silicon, real workload, real wattmeter. This is the paper's empirical anchor.
- **Level 2 (modelled):** The eco chip analytic model in Section 5. Defensible arithmetic on published component parameters. Consistency, not validation.
- **Level 3 (hypothesis):** The general design principle and its transfer to other computing layers and other domains, in Sections 3, 6 and 7. Testable in principle, not established here.

The paper's central claim — that equilibrium-based control produces the predicted efficiency gains — is demonstrated at Level 1 for one specific workload and one specific implementation. The broader claim that the principle transfers to other layers of the compute stack and to other domains is a hypothesis, and the transfer must be established by further empirical work in each specific application. The paper does not claim that a single benchmark establishes a universal theory.

Related work. Fragments of the equilibrium philosophy already exist in several places, and this paper is not the first to notice their value in isolation. The four properties defined in Section 3 map onto existing systems as follows:

System / mechanism	Sustained>peak	Predictive>reactive	Quiet default	Coordinated
Cascade / early-exit inference (Teerapittayanon 2016, Kaya 2019)	partial	no	no	no
Mobile SoC homeostatic thermal control	yes	partial	no	partial

Linux schedutil / FreeBSD powerd governor	partial	no	partial	no
Google DeepMind data centre cooling	yes	yes	no	yes
Modern web-scale autoscalers	no	partial	no	partial
Kubernetes vertical pod autoscaling	no	partial	no	no
Proposed here: equilibrium-based stack	yes	yes	yes	yes

The representative systems examined here each implement only a subset of the four properties in a specific context. None applies the philosophy consistently across the stack. The contribution of this paper is not any single property in isolation — each is a well-known technique — but the observation that they form a coherent design philosophy when applied together, and that the fragmentary partial adoption visible across the industry leaves substantial efficiency on the table that consistent application would recover.

Cascade and early-exit inference architectures (Teerapittayanon et al. 2016, Kaya et al. 2019) established that routing easy inputs to cheap paths reduces average inference cost; the contribution here is not the cascade pattern itself but the specific use of a committed, hashed governance rule as the first-stage filter, with cryptographic auditability of the routing decision. Mobile-oriented SoC designs have demonstrated homeostatic frequency and voltage control targeting sustained rather than peak operation, achieving significantly higher sustained efficiency than reactive throttling. Google's data centre cooling work using ML for predictive thermal management demonstrated approximately 40% reduction in cooling energy through equilibrium-based rather than reactive control. Modern operating system power governors (schedutil on Linux, powerd on FreeBSD) implement partial homeostatic control at the CPU frequency level. None of this work is claimed by this paper. What is claimed is that these fragments are instances of the same underlying design philosophy, and that articulating them as such — and applying the philosophy consistently across the whole stack rather than in isolated subsystems — recovers efficiency that isolated optimisation misses.

3 The principle stated

Equilibrium-based control, as used in this paper, refers to a design philosophy with four core properties.

Sustained rather than peak. The system targets a sustainable operating point rather than pushing toward maximum capability. Peak output is available when the outcome genuinely requires it, but the default is a level of operation that can be maintained without thermal stress, resource depletion, or resource

waste. Peak becomes a controlled and intentional escalation rather than the default operating mode.

Predictive rather than reactive. The system anticipates upcoming conditions using available signals (workload patterns, environmental data, scheduled events, user behaviour) and pre-positions to meet them efficiently, rather than waiting for conditions to change and then responding. Reactive control is inherently late, always chasing conditions that have already changed. Predictive control operates in the future tense, meeting conditions as they arrive rather than after they have already caused disturbance.

Quiet by default. When the drive is satisfied, the system quiets rather than continuing to maximise. A satisfied thermal target does not require continued cooling effort. A satisfied throughput target does not require additional resources to be allocated. A satisfied user request does not require the system to remain in an elevated readiness state. Quiet default means genuine deep-sleep when work is absent, rather than continued baseline consumption "just in case."

Coordinated across coupled subsystems. When multiple subsystems share resources or influence each other's operation, they coordinate rather than each locally maximising. A CPU that knows the GPU is also active can share thermal budget rather than both pushing to independent thermal limits. A cooling system that knows heating will begin shortly can pre-position rather than fight the coming demand. Coordination requires that subsystems share information and jointly optimise, rather than each operating on local information alone.

These four properties, applied consistently across a system's layers, produce a design that operates at markedly higher efficiency than the maximisation-based default.

A formal statement of the control objective. Stated as a control problem, the difference between maximisation-based and equilibrium-based control can be made precise. Given a required outcome or service level $Q_{\min}$ (throughput, latency bound, thermal envelope, accuracy target, or any combination), a maximisation-based controller seeks to maximise Q subject to hardware and operational constraints, treating resource expenditure R as a cost to be traded against Q . An equilibrium-based controller inverts this: it seeks to minimise R subject to $Q \geq Q_{\min}$, treating any output above $Q_{\min}$ as waste rather than gain. Escalation above the equilibrium operating point is permitted only when $Q_{\min}$ cannot otherwise be maintained, and returns to equilibrium as soon as it can. The two controllers can behave identically when $Q_{\min}$ is close to the hardware ceiling and the workload demands sustained peak. They diverge sharply when $Q_{\min}$ is well below the hardware ceiling or when workload is bursty, which is the case that describes most modern computing outside dedicated training runs.

The following sections present three lines of evidence for the claim that this design philosophy, so specified, produces measurable efficiency gains at every scale where it applies.

4 A measured result

Two contributions are demonstrated in the experiment below and should be distinguished explicitly, because they are separable and a reader might value one without the other.

The energy contribution comes from the cascade. Routing easy inputs to a cheap first-stage classifier and escalating only ambiguous cases to an expensive model is a well-known efficiency pattern, and the measured 64% energy reduction is attributable to this pattern rather than to anything specifically novel in the routing logic itself.

The governance contribution comes from making the cascade's decision policy committed, hashed, auditable, and independently verifiable. The first-stage rule was committed and cryptographically bound before the experiment ran; the routing decisions produced records suitable for later verification; the decisive-path outputs were independently checked against the expensive-path outputs. These are the properties (P1–P6) specified in the companion paper on the governance overlay. Without them, the cascade is an efficiency technique. With them, the cascade becomes an auditable decision procedure whose efficiency and correctness can both be verified after the fact.

The experiment measures both contributions together in a single demonstration. The energy result depends only on the cascade pattern. The governance properties are what make the result specifically an instance of the framework this paper proposes, rather than just an instance of cascade inference more generally.

A concrete demonstration of equilibrium-based control at the software layer was performed on standard silicon. The workload was sentiment classification, a common inference task suitable for benchmarking because the reference implementation (DistilBERT-base, sentiment fine-tuned) is well-characterised and the correctness of outputs is straightforward to verify.

Design. Two implementations were compared on identical input sets:

- **Baseline (B1):** DistilBERT-only. Every input classified by full transformer inference.
- **Cascade (B2):** Governance overlay as first-stage filter. Inputs matching a committed lexicon rule with high confidence are classified directly by the overlay; ambiguous inputs are escalated to DistilBERT. The lexicon rule was committed and hashed before evaluation (SHA-256 f06aea0c5ebf34609a7846fafaa20c0f92af56412415ec01ace82758751bf4b9), preventing post-hoc adjustment.

Instrumentation. CPU package power was measured using LibreHardwareMonitor via WMI polling on an Intel i7-8700 running Windows. Idle baseline was measured for 60 seconds before each run and subtracted from workload measurements. Each workload was run three times with 300 items per class (600 total items, 100% coverage across both sentiment classes after the v2 dataset fix). CPU-time energy estimates using process time × TDP were computed as a cross-check.

Results.

Workload	Wall ms/dec	CPU-time energy	LHM measured	LHM delta over idle
B1 DistilBERT alone	13.19 ± 0.11	856 mJ	897 ± 20 mJ	774 ± 21 mJ
B2 cascade	5.52 ± 0.04	359 mJ	342 ± 22 mJ	277 ± 22 mJ

Cascade / DistilBERT-alone ratio: **0.358** (measured, idle-delta), **0.425** (CPU-time estimate).

Accuracy check. The decisive items handled by the overlay were independently classified by DistilBERT to verify agreement. Positive-direction agreement: 100% (300/300). Negative-direction agreement: 100% (300/300). Overall agreement: 100% (600/600). No accuracy was traded for the energy saving.

Interpretation. On this workload, the cascade delivered the same output as monolithic DistilBERT while consuming approximately 36% of the energy per correct decision. The efficiency gain comes from the overlay's ability to handle unambiguous cases at near-zero cost without invoking the expensive model. Because agreement is 100%, the ratio measures pure efficiency rather than an accuracy-for-energy trade.

Caveats. The workload was well-matched to the lexicon rule; harder or more variable workloads will produce lower agreement rates and require the cascade to escalate more often, reducing the efficiency gain. The result is a demonstration that the principle works when the classifier is correctly calibrated for the workload, not a claim of a universal 64% saving on all inference tasks. Every deployment of the pattern must test the classifier's accuracy on its actual data distribution before the efficiency claim transfers.

The reference implementation, benchmark harness, dataset generator, and raw results are available in the accompanying repository.

5 A hardware corollary

The same principles that produced the software cascade result can be applied at the silicon layer. The eco chip architecture (specified in UK patent GB2606946.8) is one such application: a four-die chiplet stack in which a Governance Die filters decisions before they reach the compute chiplets, and a Somatic Power Rail, Constitutional Friction Circuit, and Equilibrium Drive implement homeostatic thermal management rather than reactive throttling.

An analytic energy model of this architecture was constructed to test whether the claimed 2.74 TOPS/W sustained efficiency is consistent with defensible arithmetic on published component-level parameters. The load-bearing assumption is the sustained fraction — the ratio of sustained delivered throughput to peak theoretical throughput — which conventional reactive throttling achieves at 0.50-0.70 (mid 0.60) and which proactive homeostatic control is credibly claimed to achieve at 0.70-0.88 (mid 0.80), based on published efficiency results for recent mobile-oriented SoC designs.

Result. The analytic model produces a defensible range of 2.28 to 2.87 TOPS/W with a mid-scenario of 2.61 TOPS/W. The patent's 2.74 TOPS/W claim falls inside this range, above the mid-scenario, at the top of the defensible band. The +40% advantage over a conventional silicon baseline of 2.0 TOPS/W holds across the entire range: even the low scenario (2.28 TOPS/W) beats 2.0 TOPS/W by 14%.

Single point of failure. If the Equilibrium Drive does not deliver its homeostatic advantage and the eco chip's sustained fraction collapses to conventional's 0.60, TOPS/W falls to 1.96, below the baseline. Every other input can vary by $\pm 20\%$ without changing the verdict. The sustained-fraction claim is therefore the single assumption that must be empirically validated, which requires FPGA prototyping under continuous inference load with instrumented thermals — not analytic modelling.

Positioning. The eco chip at 2.74 TOPS/W is below modern data-centre AI accelerators running sparse INT8 workloads (roughly 3.67 TOPS/W per MLPerf inference v5.1 results). The design is not intended to compete with flagship inference accelerators on peak metrics. It is intended for edge and constrained deployment where current AI silicon does not fit: drones, medical devices, environmental monitoring, developing-world AI infrastructure, assistive technology, distributed edge computing, disaster response. In that market segment, the design's efficiency advantages plus its hardware governance capabilities represent a distinct contribution.

6 Scope and transfer hypotheses

The measured result and the analytic model both concern computing. The framework itself is not specific to computing systems, and the same design philosophy plausibly applies wherever a goal-directed system operates under resource constraints with variable demand — building climate control, electric vehicle battery management, grid-scale energy management, industrial process control (cement, steel, aluminium), cybersecurity operations, and consumer electronics power management have all been suggested by earlier drafts of this work as candidates. Each of these represents a transfer hypothesis rather than an established result. The paper does not attempt to solve any of them.

Establishing whether the principle produces the predicted gains in each specific domain is a separate empirical question, appropriately answered by domain experts rather than by extrapolation from the computing result reported here. What this paper claims is that the principle is well-enough defined to be tested in each domain, and that the pattern of fragmentary partial implementations already visible across several of these fields (predictive HVAC in some smart buildings, homeostatic thermal management in some mobile SoCs, ML-driven data centre cooling) is consistent with the general framework and suggests systematic adoption could recover further efficiency currently left on the table.

7 What this paper is and is not

This paper is: a specific claim that modern computing systems have systematically adopted maximisation-based control despite the conditions that justified this choice no longer holding; a measured result showing that an equilibrium-based cascade pattern delivers 64% energy reduction on real silicon while preserving accuracy; an analytic model showing that a hardware architecture built on the same principles is consistent with a claimed 2.74 TOPS/W efficiency, subject to one load-bearing assumption (the sustained-fraction advantage of homeostatic thermal control) that requires FPGA thermal characterisation under continuous inference load to validate; and a framing that makes the principle available to engineers in other domains for application to their own systems.

This paper is not: a claim that equilibrium as a control principle is new; a claim that any specific measured saving generalises to all workloads without empirical validation; a claim that homeostatic hardware exists in production or that any specific chip design has been physically tested; a claim that the framework should be applied without engineering judgement to systems where its assumptions do not hold.

The distinction matters. Grand claims dressed up in technical language damage credibility. Modest claims about substantive things build it. What is offered here is modest: a design philosophy that already works in fragments across the industry, articulated as a coherent whole, with one measured result and one analytic result to demonstrate that the principle produces the predicted efficiency gains under specific tested conditions.

8 Limitations

The cascade result depends on classifier calibration. The 100% agreement rate in the reference implementation was achieved on a workload well-matched to the lexicon rule. On harder or more variable workloads, the classifier's error rate will be higher and the efficiency gain will diminish accordingly. Any deployment of the cascade pattern must test its classifier on the actual distribution of its production workload.

The analytic hardware model is not empirical validation. The eco chip has not been fabricated. The 2.28-2.87 TOPS/W range represents defensible arithmetic on published component parameters and cannot substitute for FPGA prototyping under continuous inference load. The single load-bearing assumption (sustained fraction 0.70-0.88) is credibly grounded in published performance of comparable mobile SoC designs but requires empirical confirmation for this specific architecture.

Adversarial environments require additional discipline. The framework was developed for stationary or slowly-varying conditions. Applications where an active adversary shapes the workload (cybersecurity, fraud detection, content moderation) require adversarial testing as a first-class design requirement, forced escalation on cases the classifier might otherwise route

cheaply, and cryptographic bindings to prevent the governance layer itself being compromised.

Distribution drift is invisible without monitoring. Cascade systems can fail silently when the workload distribution changes and the classifier's accuracy degrades without the system noticing. Periodic re-validation against the expensive path is necessary to detect drift, and this costs efficiency in exchange for safety. Systems deploying the pattern must budget for this monitoring cost.

Cultural and incentive barriers matter. The technology to implement equilibrium-based control exists and has been demonstrated in fragments across the industry. It has not been adopted as a default because industry incentives favour opportunistic maximisation: chip vendors sell more chips when workloads are inefficient, cloud providers benefit from over-provisioning, benchmarks measure peak performance. The framework does not solve these incentive problems. It only provides a coherent alternative for those who choose to implement it.

9 Conclusion

Modern computing systems often consume substantially more energy than is necessary for a given outcome, on workloads where peak-oriented operation is not required. The evidence in this paper — one measured demonstration at Level 1, one analytic model at Level 2 — is consistent with the hypothesis that this gap is not primarily a materials science problem or a fundamental physics limit, but rather a design philosophy problem: systems designed to opportunistically maximise perform worse on the specific tested workloads, thermally and energetically, than a system designed to target sustained equilibrium. Whether this hypothesis holds across the compute stack more broadly is an open empirical question, and the paper invites the further studies that would answer it.

An equilibrium-based control philosophy applied consistently across the compute stack — hardware, operating system, workload scheduling, fleet management — would produce meaningful efficiency gains at every scale. This paper presents one measured demonstration (36% of baseline energy on a real inference workload, with 100% agreement preserved) and one analytic model (a hardware architecture consistent with 2.74 TOPS/W efficiency, an improvement over 2.0 TOPS/W conventional silicon). Neither result is a proof that the principle solves every problem it might apply to. Together they demonstrate that the principle produces the predicted gains when specifically tested, and that the principle is well-enough defined to be tested by others in other domains.

The principle itself is not new. What is new is the observation that modern computing has systematically ignored it, and that adopting it deliberately across the stack would recover a substantial fraction of currently-wasted energy. The paper does not claim more than this. Engineers who recognise their own problems in the framing are invited to test whether the principle produces the predicted gains in their specific domain. Where it does, the gains compound

across the industry. Where it does not, the specific failure mode is itself useful information about the boundaries of the framework's applicability.

The immediate next step for this specific work is FPGA thermal characterisation of the eco chip's Equilibrium Drive under continuous inference load, to convert the analytic hardware result into empirical evidence. Broader-workload evaluation of the cascade pattern (adversarial inputs, distributionally shifted data, higher-difficulty tasks) is the natural companion study, and can be performed at low cost by any group with access to standard inference hardware. Both would extend the paper's Level 1 evidence base substantially. Neither requires new silicon or specialised infrastructure.

Code and reproducibility

Reference implementation of the cascade benchmark, harness scripts, dataset generator, accuracy check, and raw results:

<https://github.com/aithoughtcouncil-hattie/equilibrium-governance> (see /bench/).

Analytic eco chip model: <https://github.com/aithoughtcouncil-hattie/equilibrium-governance> (see /analytic/).

Governance protocol specification: The Missing Layer, Thought Council preprint (2026). Governance overlay for room-temperature NV registers: Governed Measurement, Thought Council preprint (2026). ICE Fusion Architecture: Thought Council preprint (2026). Framework document: ICE Theory 101, Thought Council (2 September 2026).

References

- [1] Cannon WB. The Wisdom of the Body. Norton (1932).
- [2] Wiener N. Cybernetics: or Control and Communication in the Animal and the Machine. MIT Press (1948).
- [3] Damasio AR. Descartes' Error: Emotion, Reason, and the Human Brain. Putnam (1994).
- [4] MLPerf Inference v5.1 Results. MLCommons (2025).
- [5] Google DeepMind. Machine learning for data center cooling. DeepMind blog (2016).
- [6] Aristotle. Nicomachean Ethics. Book II (c. 340 BCE).
- [7] Maclean D. The Missing Layer: Verifiable Decision Governance for High-Stakes Systems. Thought Council preprint (2026). Section 3.5 states the protocol properties (P1-P6) that the cascade benchmark demonstrates in a specific implementation.
- [8] Maclean D. ASI Series, Papers 0-5. thoughtcouncil.org (2026). Paper 3 (Equilibrium Foundation) states the underlying framework applied here to computing systems.

[9] Maclean D. ICE Theory 101. Thought Council (2 September 2026). Plain-language framing of the equilibrium principle across substrates.

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